Capstone 5 minute talk:

Slide 1: Background

- Dense stellar remnants left behind by low to intermediate mass stars
    - Bottom left region of the HR diagram is where WD are found
- DWD is a binary system with two WD

Slide 2: DWD systems

- Binary systems will produce light curves as shown in the diagram showing a few different kinds (these sources are not necessarily systems with DW)
    - With the light curve we can measure properties of the system such as eclipsing in the system (shown by the dip in the light curve) can **allow us to measure the period of the systems orbits**
- DWD are some of the **most abundant compact object systems in MW** with **short orbital periods that produce strong GW signals**
- Making them strong candidates for GW detections

Slide 3: LISA

- These sources and be used as verifications for LISA, which is a spaced based GW detector

- DWD make good sources because we can predict these strong GW signals they produce
    - Optical surveys tell us their **position and orbital periods allowing for predictions of these signals**
- Cataloged systems with short period binaries can be detectable with in 3 months of LISA operation allowing for calibration for LISA to then detect other compact systems such as BH-BH, BH-WD, BH-NS, NS-NS, and NS-WD

Slide 4:

- When using a specialized light curve synthesis and modeling code to analyze photometric observations these systems fall into distinguishable categories based on their modulations
- Top J0651
    - Full eclipsing system with period of 12.75 minutes shown through the deep v shapes from one passing in front of the other
    - **Precise eclipsing timing can pin down inclination measurements and the masses of the secondary and primary sources** in the system
        - There is also some doppler and ellipsoidal modulations as shown in the light curve at the top of the arcs

- Bottom J0106
    - Has a smooth shape with no sharp drops this shape is caused by tidal stretching which is an ellipsoidal system
        - tidal modulations are essentialling the stretching of the stars caused by the gravity of the companion
    - **Without eclipsing we lose certain geometric edge timing leading to more mass uncertainty that relies on spectroscopic priors** (such as the parameters made with electromagnetic measurements)

Slide 5: Featureless

- Lastly J2322
    - A featureless system which are a great challenge with the massive error bars
        - **The models essentially make a light curve of best fit which for this system is just a straight line**
    - This **means that there are no observable modulations which leaves the orbital inclination measurements being almost entirely uncertain**
        - This system has inclination uncertainty of ~84%

    - These systems have severe mathematical degeneracy as a result, which **inflates the error on the secondary mass and weakening in the GW signal prediction**

Slide 6: How do we characterize these Featureless systems

- J2322 is unfortunately not an isolated system because **these systems make up ~54% of all DWD candidates found**
- For these systems when **observed in the UV wavelength the secondary source is drowned out by the primary source**
- So the question is how do we correct for this
    - We believe we can look at the systems in the **lower inferred wavelength where the secondary source should have more influence over the light curve**

Slide 7: My project

- So my project is looking at these candidates in the inferred
    - So we can **apply the spectroscopic priors to the system to correct for the degeneracy found in the UV**
- Thus far i have only looked at 14 sources and identifying which will be **better candidates to observe in the inferred by looking for sources with not as much noise caused by the systems neighbors**

- J0935 system
    - So far the only source that i have been able to get a light curve in the inferred
    - I have not analyzed the light curve just yet because i am trying to get light curves for some of the other sources i am looking at

Slide 8: summary

- So the main limitation i am looking at is that the **featureless systems make up ~54% of potential validation candidates** for LISA detection
- To solve this we are looking at the sources in the **inferred to see if we can lower the degeneracy by applying priors to the secondary source in addition to the primary source in the UV**
- With these tighter constraints for both the secondary and primary source we can **predict more price GW signals to help calibrate LISA**